

Personal Project
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Design Project: Inline 4-Cylinder Toy Engine

Personal Design Report

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1 Introduction

The purpose of this project was to develop my fundamental knowledge of the underlying mechanisms involved in the typical *Internal Combustion Engine* (ICE), While also showcasing my skills and understanding of Autodesk Inventor and Autodesk Fusion 360.

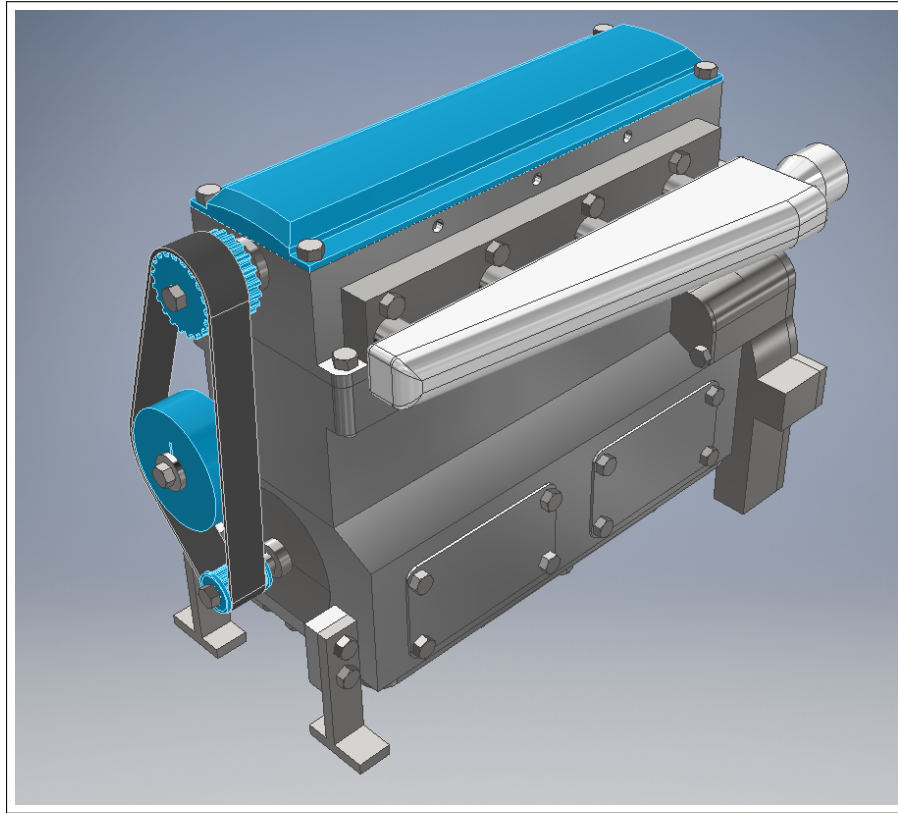


Figure 1: Final Design

Various tools used in the project include:

- Surface Modelling
- Assembly Modelling
- Gear Creation, Gear trains, BASIC belt drive, and Assembly Motion Mechanisms
- CAM

2 Design

2.1 The Internal Combustion Engine

Through some thorough research, I have learned the basics as to how the basic engine works:

- The engine is turned on via a starter motor that is connected to a gear called the flywheel. This flywheel is connected directly to the crankshaft, and its initial moment turns the crankshaft to begin the basic motion.
- The crankshaft is a rod that contains 4 piston rods which connect to pistons. The crankshaft transfers rotational motion to the linear motion of the piston had within the cylinders that they reside in.
- The crankshaft is also directly connected to the camshaft via synchronous belt drive. The camshaft is a rod containing cams that sit above the valves (intake valve for releasing air and fuel, and exhaust valve for removing the combustion exhaust) that are connected on the cylinder. The camshaft rotates at have the speed of the crankshaft.
- This entire system is a repetitive 4-cycle, in which each stroke (180° turn of the crankshaft) completes a specific function:
 - First stroke: the *intake stroke*, the intake valve opens up in the cylinder and the piston moves downward, allowing air/fuel to be released into the cylinder.
 - Second stroke: the *compression stroke*, the intake valve closes and the piston moves up, compressing the contents in the cylinder.
 - Third stroke: the *power stroke*, the spark plug located at the top of the cylinder ignites the compressed contents of the cylinder, creating enough power (through combustion) to push the cylinder downwards.
 - Fourth stroke: the *exhaust stroke*, the cylinder moves upward again, and the exhaust valve opens, allowing the byproducts of the reaction to be pushed out of the exhaust hole and out of the engine.

With all of this in mind, I plan to make my engine following these concepts, without some of the extra details that would over complicate the project for now.

2.2 Design Choices

Since this is my first preliminary design of the engine that showcases my understanding of the main mechanisms, I have decided to not include wiring of the spark plug, or coolant setup since they also cannot be simulated in motion, but are also not translated well into Autodesk Inventor. It is also difficult to actually simulated the combustion reaction actually occuring, so I have just included the spark plug in the general area where it would be.

2.3 Initial Design

I went through the design process to draft an original design, in which I used that design to make improvements and draft the final design that can be seen in figure 1. The initial design ran into several issues, such as sizing of the gears, lack of fixtures between parts, pistons that were oversized, and several other smaller issues. These were then improved in the final design, which can be seen in the comparison figure 2 below.

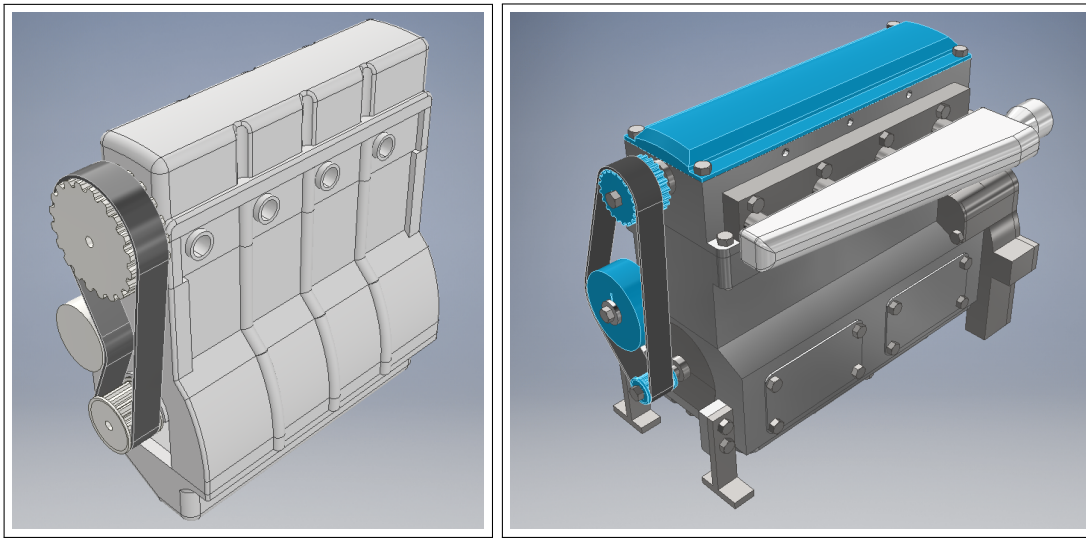


Figure 2: Comparison of engines

As, we can see, the design is not very feasible for creation, and does not rightfully simulate the engine design I was intending on.

2.4 Debugging and Optimizations

One of the biggest issues in my first design was the lack of implementation of bolted connections so that the build could actually be manufacturable. Along with this, designing the gears to be realistic to the size of the rest of the engine, while also implementing the intake and exhaust manifolds so that the model could be more accurate.

There were several issues with the sizing of various parts, like the engine block and cylinder head, which caused issues because of a lack of good pre planning. I designed my engine block to be thicker to account for the various add-ons and sections present in the design.

2.5 Future Optimizations

After designing the new engine, I had encountered more issues in my design that I plan to use in future designs. One common issues is to design the gears first prior to making the engine block and cylinder head. This would allow better sizing and form of the overall design. I noticed that my design choice of using SOHC (single overhead camshaft), which was mainly in place since the gears and engine block/cylinder head were sized incorrectly. In future designs, I would rather draft my engine to hold two sets of gears that control the camshafts and use DOHC (double overhead camshaft) so that they can be accurate to actual engines. Along with this, I am lacking the space in my cylinder head to incorporate taps that would be underneath the camshaft and be pressing on the valves.

Highlighting failures in my current design reinforce the cyclical design process involved in engineering and allow me to improve in the future.

3 Simulations

3.1 Motion Study

A motion study was completed showcasing the flywheel-crankshaft-belt-camshaft connection, while also ensuring the valve timing matches correctly in the 1-3-4-2 firing order that is standard for the

inline 4-cylinder engine.

3.2 CAM Simulation

In order to simulate actually manufacturing the piece, I completed a CAM simulation on Fusion 360. Using the *Kurtz Ersu Alpha 140* 3D printer, which uses metal powder bed fusion (MPBF) printing. The material used was *stainless steel 1.4404-316L 90*. The idea of the MPBF printer is to utilize several layers of metal powder, in which a high temperature laser or electron beam is used to fuse the powder together on a particle level to create a strong bond. This form of printing creates highly detailed complex forms, while also reducing material waste since it is not a material removal related process.

I generated the tool path and print time of the engine block, the most complex form in the design. It can be seen in figure 3.

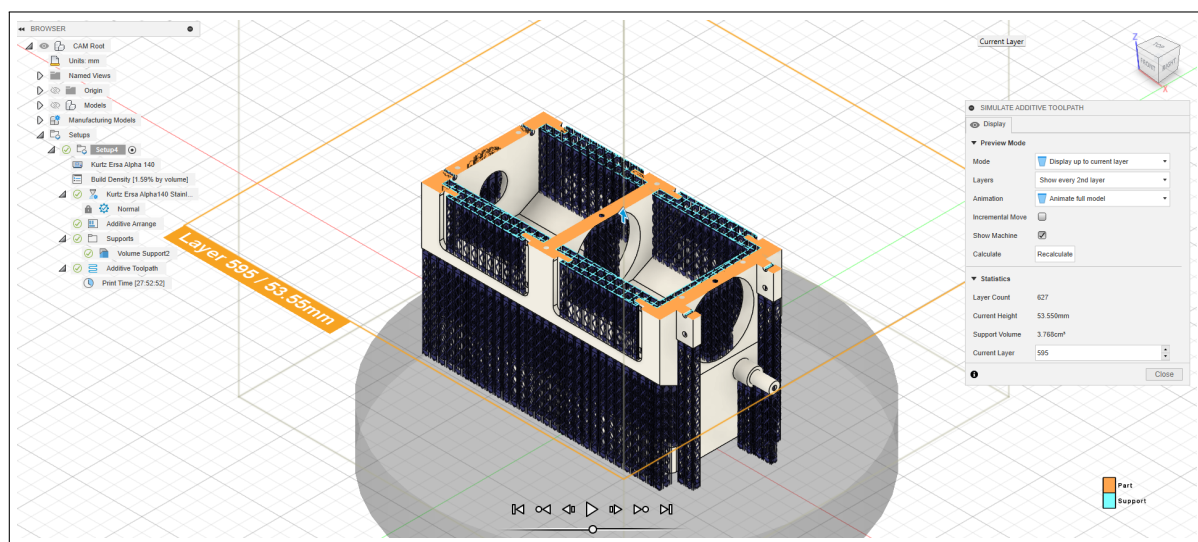


Figure 3: CAM simulation

As noted in 3, the print time is roughly 25 hours, which makes sense when considering the complexity of the form.

In future designs I plan on simplifying the design in some way, through either splitting the parts of the engine block or just by reducing the the overall complexity of the shape so that the shape can be printed in a much faster time, since it cannot be realistically manufactured with 25 hour print times for a single piece.